

Lesson-1 : Introduction to Real time and Embedded System

SEng3441: Real time and Embedded Systems

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Lesson-1 Outline

- What is System?
- What is an Embedded System ?
- What is Real Time?
- ES Application
- RT Application
- Real Time and Embedded System
- Some common RTES Application
- Computer organization and ES concept
- Architecture of an Embedded System

Definitions RT and ES

What is System?

- A **System** is a way of working, organizing or performing one or many tasks according to a fixed set of rules, program or plan.
- Examples:
 - Time display system- watch
 - Automatic cloth washing system –A washing machine

What is an Embedded System ?

- **An embedded system** is a system that has software embedded into computer-hardware, which makes a system dedicated for an application (s) or specific part of an application or product or part of a larger system

What is Real Time?

- **Real-Time:** refers to the ability of a system or process to respond to inputs, events, or stimuli and produce outputs or actions within a specific and predefined time frame.

Key Characteristics of Embedded Systems:

- **Task-Specific:** Designed to perform a particular function or set of functions
- **Real-Time Operation:** Often need to process data and respond to inputs in real-time.
- **Resource Constraints:** Limited processing power, memory, and storage compared to general-purpose systems.
- **Integration:** Embedded within larger systems, such as appliances, vehicles, or industrial equipment.
- **Reliability:** Designed to operate consistently for long periods, often in challenging environments.

A Real-Time Embedded System

- **A Real-Time Embedded System:** is an embedded system designed to perform specific tasks within a strict time constraint.
- These systems must produce predictable and deterministic outputs in response to specific inputs or events, often in critical applications where timing is as important as correctness.

Key Characteristics of Real-Time Embedded Systems:

- **Timeliness:** Tasks must be completed within defined deadlines.
- **Determinism:** The system's behaviour is predictable and repeatable.
- **Reliability:** Highly dependable to prevent catastrophic failures in critical applications.
- **Minimal Latency:** Response times are short and consistent

Examples of Embedded Systems:

Examples of Embedded Systems:

- **Consumer Electronics:** Smartphones, smart watches, digital cameras.
- **Automotive:** Anti-lock braking systems (ABS), engine control units (ECUs), airbags.
- **Home Appliances:** Washing machines, microwaves, refrigerators.
- **Industrial Applications:** Robotics, process control systems.
- **Medical Devices:** Pacemakers, infusion pumps, medical imaging systems.

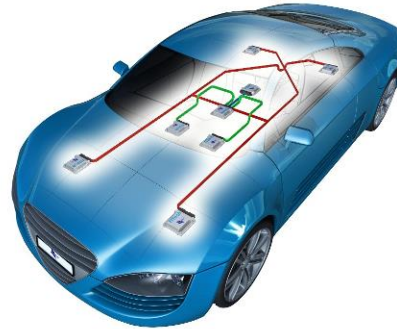
Definitions RT and ES

ES Application



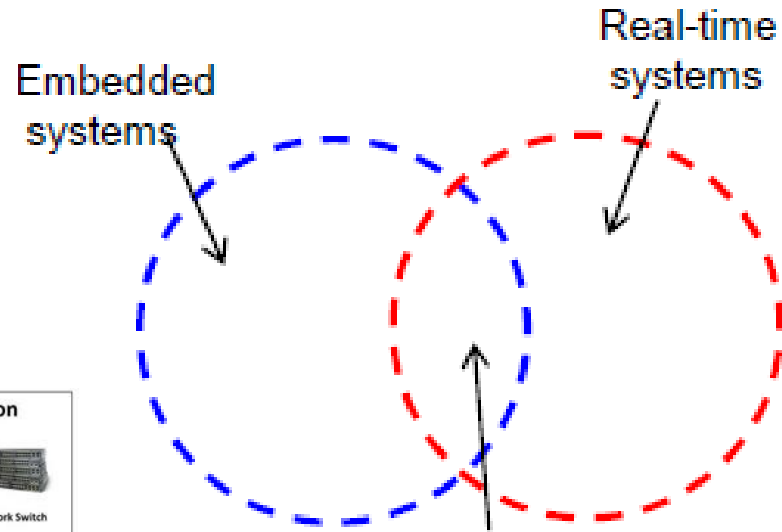
Definitions RT and ES

RT Application



Definitions RT and ES

RT and ES



Some common RTEs Application

- Flight control
- Space missions
- Railway switching systems
- Military systems
- Plant control
- Automotive applications
- Robotics
- Household appliances,
- Virtual / augmented reality, etc.

Computer organization and ES concept

- **Computer Organization:** Computer organization deals with the operational structure and functional aspects of a computer system.
- It focuses on how hardware components interact and how a computer executes instructions.

Key components in general purpose computer :

1. Basic Building Blocks:

- **Processor (CPU):** The brain of the computer; it performs arithmetic, logical, control, and input/output operations.
- **Memory:**
 - **Primary Memory (RAM, ROM):** Fast, volatile, or non-volatile memory used for immediate processing.
 - **Secondary Memory:** Storage devices like SSDs and HDDs.

Computer organization and ES concept

- **Input/output (I/O) Devices:** Hardware for interacting with the system (keyboard, mouse, display).
- **Buses:** Communication pathways connecting CPU, memory, and peripherals

2. Instruction Set Architecture (ISA):

- Defines the set of instructions a CPU can execute.
- Examples: RISC (Reduced Instruction Set Computing) and CISC (Complex Instruction Set Computing).

3. performance Metrics:

- **Clock Speed:** Determines how fast the CPU executes instructions.
- **Throughput:** Number of instructions or processes executed per unit time.
- **Latency:** Time taken to complete a task.

Computer organization and ES concept

4. **Cache Memory:** Small, high-speed memory located near the CPU to store frequently accessed data.

5. **Pipelining:** Technique where multiple instruction stages (fetch, decode, execute) are processed simultaneously for efficiency.

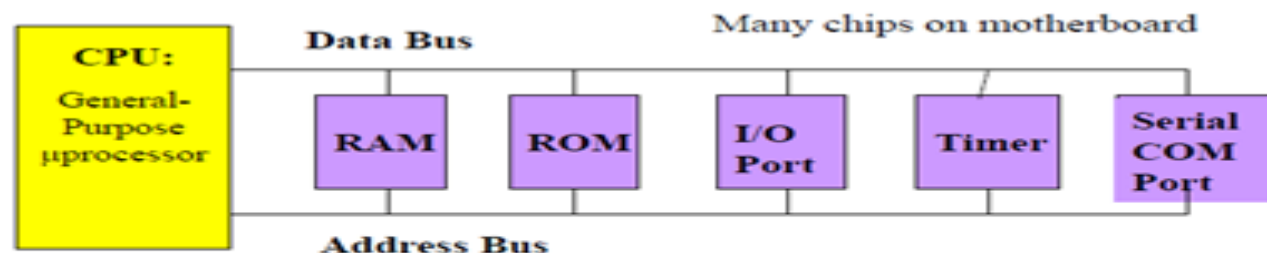
5. **Parallelism:** Multiple processors or cores work together to execute tasks.

Computer organization and ES concept

Components of A General Purpose Computer

General-purpose Microprocessor:

- CPU for Computers
- Commonly no RAM, ROM, I/O on CPU chip itself



Computer organization and ES concept

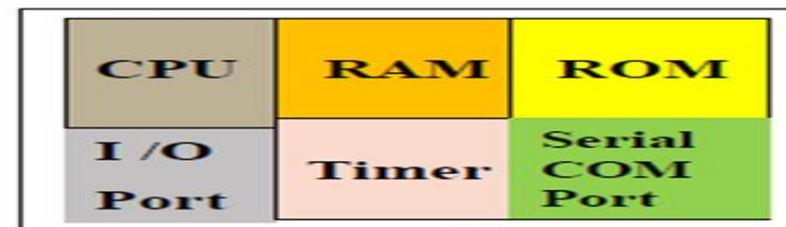
- Embedded systems are specialized computing systems designed to perform dedicated tasks, often in real-time environments.
- They are integrated into larger systems to control functions and operations.

Key Characteristics of Embedded Systems:

- **Purpose-Specific:**
 - Designed for a specific function or application.
 - Example: A microcontroller in a washing machine.
- **Real-Time Operation:**
 - **Hard Real-Time:** Must meet strict deadlines (e.g., medical equipment).
 - **Soft Real-Time:** Flexible deadlines (e.g., video playback).

Computer organization and ES concept

- Hardware and Software Integration: Typically includes microcontrollers or microprocessors.
- Uses embedded software (firmware) programmed to perform tasks.
- **Resource Constraints:** Limited memory, processing power, and energy consumption.
- **Reliability and Safety:** Designed to operate continuously without failure in critical applications (e.g., automotive systems, avionics).
- Example : Motorola's 6811, Intel's 8051, Atmel 128, Zilog's Z8 and PIC 16X, Atmel328/ Arduino et



A single chip
microcontroller

Computer organization and ES concept

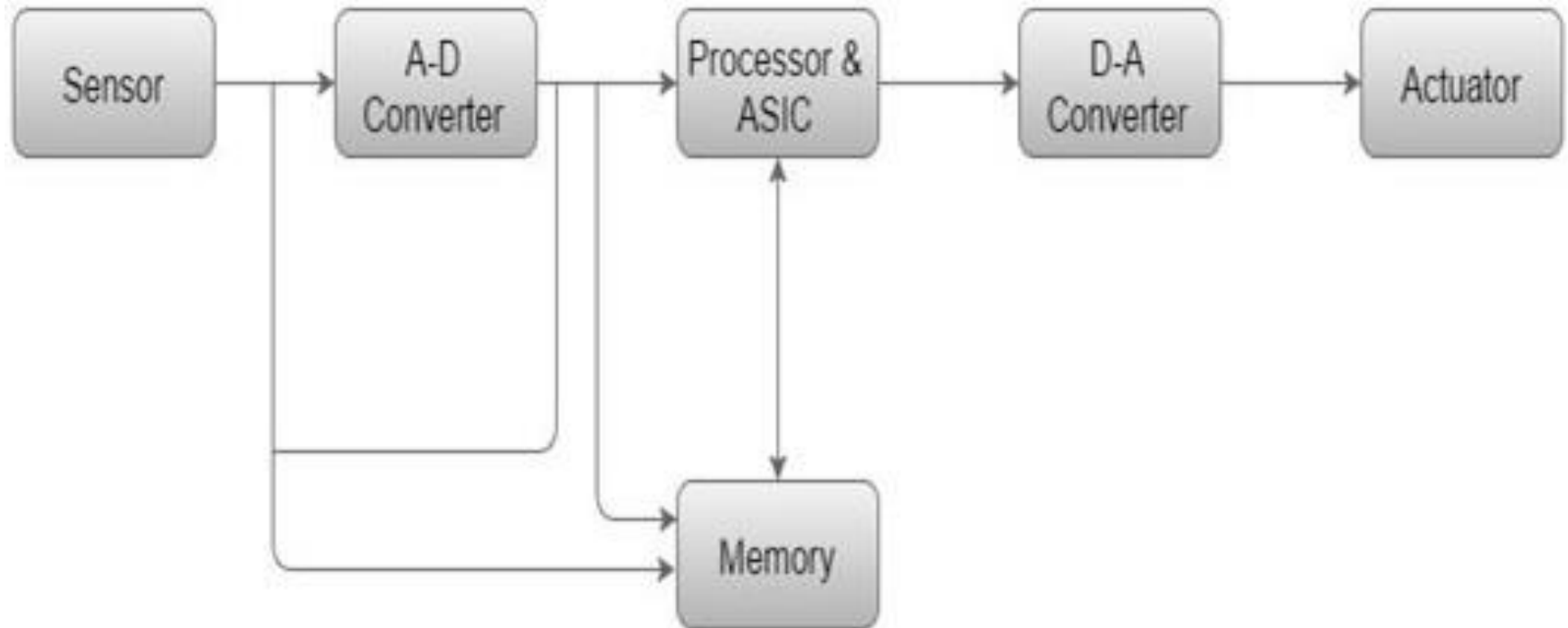
Components of an Embedded System

Microcontroller

- A single-chip computer
- On-chip RAM, ROM, I/O ports...

Computer organization and ES concept

Architecture of an Embedded System



- End of lesson One